

<page\_start no = 1>

<sol\_start id=1>

1. (3)

<sol\_end>

<sol\_start id=2>

2. No solution

<sol\_end>

<sol\_start id=3>

3. 10 units

<sol\_end>

<sol\_start id=4>

4. Proportional

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<sol\_start id=5>

5.  $\frac{-1}{2}$

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<sol\_start id=6>

6.  $x^3 y^3$

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<sol\_start id=7>

7. Sem of zeroes =  $\frac{1}{2}$  their product

$$\alpha + \beta = \frac{1}{2} \alpha\beta$$

$$\frac{-b}{a} = \frac{1}{2} \left( \frac{c}{a} \right)$$

$$\frac{1}{2} = \frac{k-4}{8}$$

$$8 = 2k - 8$$

$$2k = 16$$

$$k = 8$$

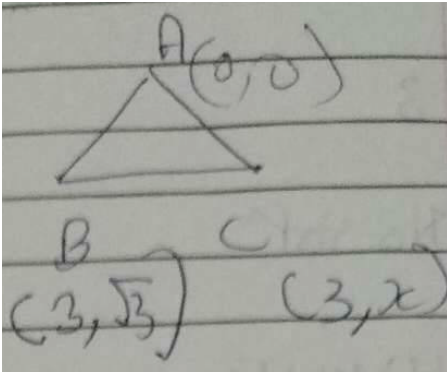
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<sol\_start id=8>

8  $AB^2 = AC^2$  (because it can equable al  $\Delta^{1e}$ )



$$(3-0)^2 + (\sqrt{3}-0)^2 = (3-0)^2 + (x-0)^2$$

$$\Rightarrow (\sqrt{3}-0)^2 = (x-0)^2$$

$$x = \sqrt{3}$$

<sol\_end>

<sol\_start id=9>

9.  $2px + 3y = 7$

$$2x + y = 0$$

$$\frac{2p}{x} \neq \frac{3}{1} \quad (\text{Given})$$

$$2P \neq 6$$

$$P \neq 6$$

<sol\_end>

<sol\_start id=10>

10.  $PD \parallel BC$

$$AP/PB = AD/DC \quad (\text{B.P.T}) \quad \rightarrow (1)$$

$$AD/DC = CE/BE \quad (\text{Given}) \quad \rightarrow (2)$$

$$\frac{AP}{PB} = \frac{CE}{BE} \quad (\text{From (1) \& (2)})$$

by converse of BPT

$$DE \parallel AC$$

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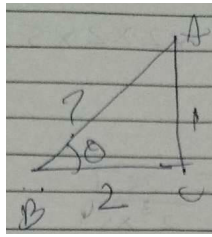
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11.  $\tan \theta = \frac{\text{opp}}{\text{ady}}$

$$AB^2 = AC^2 + BC^2$$
$$= 1 + 4$$
$$= 5$$
$$AB = \sqrt{5}$$



$$\sin \theta = \frac{\text{app}}{\text{hypotenuse}}$$
$$= \frac{1}{\sqrt{5}}$$

$$\cos \theta = \frac{\text{adj}}{\text{hypotenuse}}$$
$$\frac{1}{2} \sin \theta \cos \theta = \frac{1}{5}$$

$$\frac{1}{2} \times \frac{1}{\sqrt{5}} \times \frac{2}{\sqrt{5}} = \frac{1}{5}$$

$$\frac{1}{5} = \frac{1}{5} \text{ Hence proved}$$

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<sol\_start id=12>

12.  $5000 = 2 \times 2 \times 2 \times 5 \times 5 \times 5 \times 5$

$$\frac{257}{5000} = \frac{257}{2^3 \times 5^4}$$

$$\Rightarrow \frac{257}{2^3 \times 5^4} \times \frac{2}{2}$$

$$\frac{514}{(10)^4}$$

$$= 0.0514$$

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<sol\_start id=13>

13.  $\sqrt{3} = \frac{p}{q}$

let is first assure that

$\sqrt{3}$  is rational and is the form  $p/q$  and  $q \neq 0$

( $p/q$  are co-prime integers)

Square on both sides

$$3 = \frac{p^2}{q^2}$$

$$3q^2 = p^2$$

$\Rightarrow 3$  is a factor of  $p^2$

3 is a factor of  $p$  also

$$p = 3k$$

square on both sides

$$p^2 = 9k^2$$

$$3q^2 = 9k^2$$

$$q^2 = 3k^2$$

$\Rightarrow 3$  is a factor of  $q^2$

$\therefore 3$  is a factor of  $q$  also

3 is now the factor of both  $p$  and  $q$

This contradicts our statement that  $p/q$  are co-prime integers

$\therefore \sqrt{3}$  is an irrational us

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<sol\_start id=14>

$$14. \quad \alpha + \beta = \frac{-3}{2}$$

$$\alpha\beta = \frac{7}{2}$$

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} \Rightarrow \frac{\alpha^2 + \beta^2}{\alpha^2\beta^2}$$

$$= \frac{\left(\frac{-3}{2}\right)^2 - \left(2\frac{7}{2}\right)^2}{\left(\frac{7}{2}\right)^2}$$

$$= \frac{\frac{9}{4} - 7}{\frac{49}{4}}$$

$$= \frac{-18}{49}$$

$$\text{Also } \frac{1}{(\alpha\beta)^2} = \frac{4}{49}$$

$$x^2 + \left(\frac{18}{49}\right)x + \frac{4}{49}$$

$$49x^2 + 18x + 4 = 0$$

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<sol\_start id=15>

15. 
$$\frac{3}{5} = \frac{-a-1}{1-2a}$$
$$-5(a+1) = 3(1-2a)$$
$$a = 8$$
$$\frac{3}{5} = \frac{2b-1}{3b}$$
$$9b = 10b - 5$$
$$b = 5$$

<sol\_end>

<sol\_start id=16>

16. Given  $\triangle ABE \cong \triangle ACD$   
 $AB = AC$  (C. PCT)  $\rightarrow (1)$   
 $AE = AP$  (CP. CT)  $\rightarrow (2)$   
Dividing (2) by (1)  
$$\frac{AD}{AB} = \frac{AF}{AC}$$
  
In  $\triangle ADE$  &  $\triangle ABC$   
 $\angle A = \angle A$  (coupon)  
$$\frac{AD}{AB} = \frac{AF}{AC}$$
  
(proved)  
By SAS criterion  
 $\triangle ADE \sim \triangle ABC$   
Hence proved

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<sol\_start id=17>

17. 
$$\frac{\frac{\sin^2 \theta}{\cos^2 \theta}}{\frac{\sin^2 \theta}{\cos^2 \theta} - 1} + \frac{\frac{1}{\sin^2 \theta}}{\frac{1}{\cos^2 \theta} - \frac{1}{\sin^2 \theta}}$$
$$\frac{\frac{\sin^2 \theta}{\cos^2 \theta}}{\frac{\sin^2 \theta - \cos^2 \theta}{\cos^2 \theta}} + \frac{\frac{1}{\sin^2 \theta}}{\frac{1}{\cos^2 \theta} - \frac{1}{\sin^2 \theta}}$$
$$\frac{\sin^2 \theta}{\cos^2 \theta} \times \frac{\cos^2 \theta}{\sin^2 \theta - \cos^2 \theta} + \frac{\frac{1}{\sin^2 \theta}}{\frac{\sin^2 \theta - \cos^2 \theta}{\sin^2 \cos^2 \theta}}$$

$$\frac{\sin^2 \theta}{\sin^2 \theta - \cos^2 \theta} + \cancel{\cos^2} \frac{\cancel{\sin^2 \theta} \cos^2 \theta}{\cancel{\sin^2 \theta} - (\sin^2 \theta \cos^2 \theta)}$$

$$\frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta - \cos^2 \theta} = \frac{1}{\sin^2 \theta - \cos^2 \theta}$$

= RHS

Hence proved

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<sol\_start id=19>

19. Let the fraction be  $\frac{x}{y}$

$$\frac{2x}{y+1} = 1 \quad (\text{Given})$$

$$2x = y + 1$$

$$\frac{x+4}{2y} = \frac{1}{2} \Rightarrow x+4 = y$$

$$2x - y - 1 = 0$$

$$x - y + 4 = 0$$

$$x = -5$$

$$x + \cancel{5} + \cancel{4} = 0 \quad -\cancel{5} - y + \cancel{4} = 0$$

$$\cancel{-1}$$

$$y = -9$$

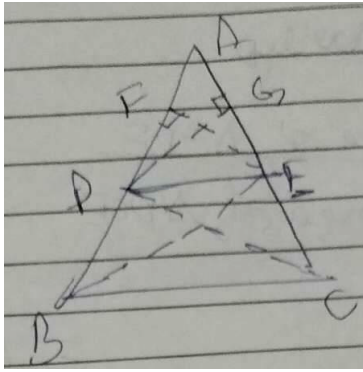
$$\therefore \text{The fraction is } \frac{5}{9}$$

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<sol\_start id=20>

20. BPT start that, if a line parallel to a side of a triangle it intersects the other 2 sides than the line divides it unto equal proportion

ABC be a  $\Delta$  i.e



$BC \parallel DE$

To prove  $\frac{AD}{DB} = \frac{AE}{EC}$

Join BE, CD

Construction :  $EF \perp AB$ ,  $DG \perp CA$

$\therefore$  Area of  $\triangle ADE = \frac{1}{2} \times \text{base} \times \text{height}$

$$= \frac{1}{2} \times AD \times EF$$

$$\text{Area of } \triangle ADE = \frac{1}{2} \times AE \times DG$$

$$\frac{\text{Area of } \triangle ADE}{\text{Area of } \triangle DBE} = \frac{AD}{DB} \rightarrow (1)$$

Similarly

$$\frac{\text{Area of } \triangle ADE}{\text{Area of } \triangle DCE} = \frac{\frac{1}{2} \times AE \times DG}{\frac{1}{2} \times EC \times DG}$$

$$= \frac{AE}{EC}$$

$\triangle DBE$  and  $\triangle DCE$  have the same base b'w same parallels

$$\therefore \frac{AD}{DB} = \frac{AE}{EC}$$

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<sol\_start id=21>

$$21. \frac{\frac{1}{2} + 2\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{\sqrt{3}}{1}\right)^2}{\frac{1}{2}\left(\frac{1}{1}\right) + \left(\frac{\sqrt{3}}{2}\right)^2 + 1^2}$$

$$\frac{\frac{1}{2} + \frac{2}{2} + 3}{\frac{1}{2} + \frac{3}{4} + 1}$$

$$\frac{\frac{3}{2} + 3}{\frac{1}{2} + \frac{3}{4} + 1}$$

$$\frac{3+6}{2}$$

$$\frac{3+6}{2}$$

$$\frac{3+6}{2}$$

$$\frac{3+6}{2}$$

$$\frac{2+3+4}{4} \Rightarrow \frac{\cancel{2}}{2} \times \frac{\cancel{4}}{\cancel{2}} = \frac{2}{1}$$

$$\therefore \frac{\sin 30 + 2 \cos^2 45 + \tan^2 60}{\frac{1}{2} \cot 45^\circ + \cos^2 30 + \tan^2 45} = 2$$

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<sol\_start id=23>

23. Time (one round) covered by =  $\frac{1980}{330} = 6$  min

Time (one round) covered by B =  $\frac{1980}{198} = 10$  min

Time (one round) covered by C =  $\frac{1980}{198} = 9$  min

A,B,C meet after time t and completed  $n_1, n_2, n_3$  rounds respectively

$$330t = x + 1980n_1 \quad \text{---(1)}$$

$$198t = x + 1980n_2 \quad \text{----(2)}$$

Substituting x into (2) & (3)

$$198t = 330t - 1980n_1 + 1980n_2$$

$$1980(n_1 - n_2) = 132t$$

$$t = 15(n_1 - n_2)$$

$$\text{Also, } 220t = 330t - 1980n_1 + 1980n_2$$

$$1980(n_1 - n_3) = 110t$$

$$t = (n_1 - n_3)$$

Since t is same and  $n_1 - n_3$  and  $n_1 - n_2$  can be whole no.

LCM of 15 and 15 and 18 is 90

A,B,C will meet after 90 mins

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<sol\_start id=24>

24.  $(2y + x = 70) \times 2$  Father's age be  $x$  Son's age be  $y$

$$22 + y = 95$$

$$4y + 2x = 140$$

$$y + 2x = 95$$

$$\Rightarrow 3y = 45$$

$$y = 15$$

$$15 + 2x = 95$$

$$2x = 80$$

$$x = 40$$

Father's age 40

Son's age 15

<sol\_end>

<sol\_start id=25>

25.  $(6,4)$  and  $(1,-7)$  be divided by the point on  $x$  axis  $(x,0)$

Let us assume the ratio to be  $k : 1$

$$\frac{-7k + 4}{k + 1} = 0$$

(section formula)

$$-7k + 4 = 0$$

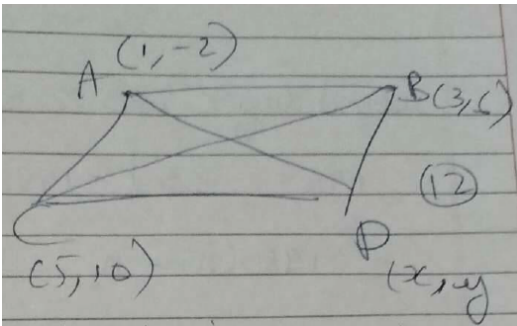
$$k = \frac{4}{7}$$

$$m : n \Rightarrow \frac{4}{7} : 1 \Rightarrow 4 : 7$$

$$m + n = 4 + 7 = 11$$

W.K.T

Diagonals of a  $\parallel^{\text{gm}}$  bisect each other.



So the midpoint of AC is same of mid point of BD

AC

BD

$$\left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) = \left( \frac{3 + x}{2}, \frac{6 + y}{2} \right)$$

$$\left( \frac{3 + x}{2}, \frac{6 + y}{2} \right) = \left( \frac{3 + x}{2}, \frac{6 + y}{2} \right)$$

$$\frac{6}{2}, \frac{8}{2} = \frac{3 + x}{2}, \frac{6 + y}{2}$$

$$3 + x = 6 ; 6 + y = 3$$

$$x = 3 ; y = 2$$

So forth vertex's coordinates are (3,2)

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<sol\_start id=28>

$$28. \quad D \text{ of PR} = \frac{8-2}{2}, \frac{3-5}{2}$$

$$= \frac{6}{2}, \frac{-2}{2}$$

$$= 3, -1$$

<sol\_end>

<sol\_start id=29>

29. Relative speed  $x - y$  (same direction)

$$\text{Dist} = (x - y)5$$

$$(100)$$

$$x - y = 20$$

$x + y$  (oppilate direction)

$$x + y = 100$$

$$x - y = 20$$

$$\underline{x - y = 100}$$

$$2x = 120$$

$$x = 60 \text{ km/h}$$

$$y = 40 \text{ km/h}$$

<sol\_end>

<sol\_start id=30>

30.  $\triangle ABD \sim \triangle PQD$  (corresponding sides are parallel)

$$\frac{x}{Z} = \frac{BD}{QD} \Rightarrow \frac{1}{x} = \frac{QD}{Z \times BD} \quad \text{---(1)}$$

$\triangle CDB \sim \triangle PQB$  (corresponding sides are parallel)

$$\frac{BD}{DQ} \Rightarrow \frac{1}{y} = \frac{BQ}{Z \times BD} \quad \text{---(2)}$$

$$\frac{1}{x} + \frac{1}{y} = \frac{BQ + QD}{Z + BD} = \frac{x}{Z}$$

Hence prover

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